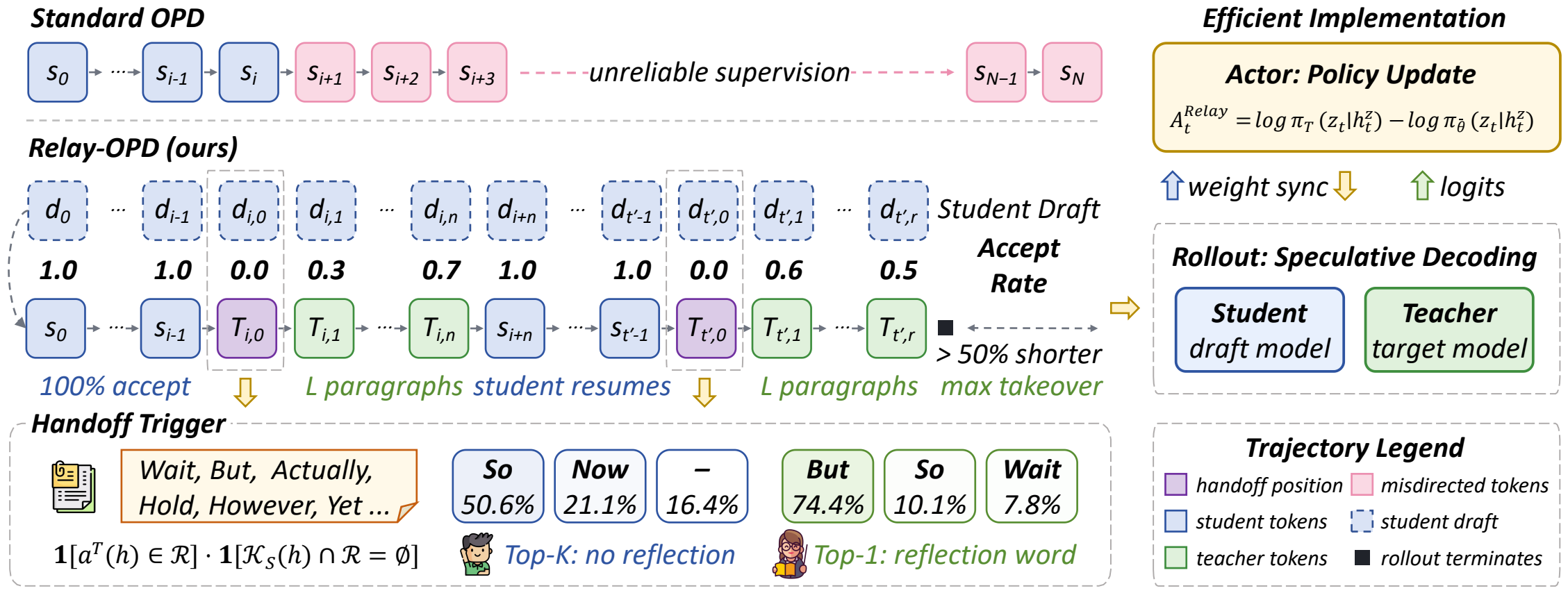

Standard OPD
s_0 s_{i-1} s_i s_{i+1} s_{i+2} s_{i+3} unreliable supervision s_{N-1} s_N
Relay-OPD (ours)
d_0 d_{i-1} $d_{i,0}$ $d_{i,1}$ $d_{i,n}$ d_{i+n} $d_{t'-1}$ $d_{t',0}$ $d_{t',1}$ $d_{t',r}$ Student Draft
1.0 1.0 0.0 0.3 0.7 1.0 1.0 0.0 0.6 0.5
Accept Rate
s_0 s_{i-1} $T_{i,0}$ $T_{i,1}$ $T_{i,n}$ s_{i+n} $s_{t'-1}$ $T_{t',0}$ $T_{t',1}$ $T_{t',r}$
> 50% shorter
100% accept
L paragraphs student resumes
L paragraphs
max takeover
Handoff Trigger
Wait, But, Actually, Hold, However, Yet ...
$\mathbf{1}[a^T(h) \in \mathcal{R}] \cdot \mathbf{1}[\mathcal{K}_S(h) \cap \mathcal{R} = \emptyset]$
So 50.6%
Now 21.1%
– 16.4%
Top-K: no reflection
But 74.4%
So 10.1%
Wait 7.8%
Top-1: reflection word
Efficient Implementation
Actor: Policy Update
$A_t^{Relay} = \log \pi_T(z_t|h_t^z) - \log \pi_{\tilde{\theta}}(z_t|h_t^z)$
weight sync
logits
Rollout: Speculative Decoding
Student draft model
Teacher target model
Trajectory Legend
handoff position
misdirected tokens
student tokens
student draft
teacher tokens
rollout terminates